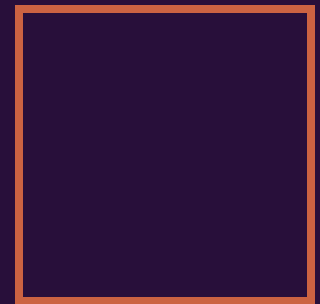


— LEA LABS · WEB DEVELOPMENT

HTML Foundations

Structure Before Style

MODULE 01



HTML gives content meaning

HyperText Markup Language (HTML) is not a programming language. It is a system for annotating text so a browser understands its structure.

■ Structure before style

HTML tells the browser what the content *is* (a heading, a list), not what it *looks like*.

■ The foundation of accessibility

Screen readers and search engines rely entirely on correct HTML to understand your page.

UNSTRUCTURED TEXT

Campus Connect

Welcome to the student portal.

Upcoming Events

- Tech Workshop
- Career Fair

STRUCTURED HTML

```
<h1>Campus Connect</h1>
```

```
<p>Welcome to the student portal.</p>
```

```
<h2>Upcoming Events</h2>
```

```
<ul>
```

```
  <li>Tech Workshop</li>
```

```
  <li>Career Fair</li>
```

```
</ul>
```

A page begins with a valid document

<!DOCTYPE html>

Required at the very top. Tells the browser to use modern HTML5 standards and avoid quirks mode.

<html>

The root element wrapping all content; usually includes a lang attribute.

<head>

The invisible brain: metadata, the page title, and links to CSS or scripts.

<body>

The visible heart: everything the user sees and interacts with.

```
<!-- 1. Document Type -->
```

```
<!DOCTYPE html>
```

```
<!-- 2. Root Element -->
```

```
<html lang="en">
```

```
<!-- 3. Metadata -->
```

```
<head> <meta charset="UTF-8"> <title>Campus  
Connect</title> </head>
```

```
<!-- 4. Content -->
```

```
<body> <h1>Welcome to Campus</h1> <p>Find  
your next event.</p> </body>
```

```
</html>
```

Elements wrap meaning around content

<p>

OPENING TAG

Marks exactly where the element begins to take effect.

**Campus
Connect**

CONTENT

The actual text or nested elements being wrapped.

</p>

CLOSING TAG

Marks where the element ends. Note the forward slash.

Void Elements

Some elements cannot contain content. They stand alone and do not use a closing tag.

**
**

<hr>

Attributes add essential detail

Attributes provide extra information about elements. They always go in the opening tag and usually come in name/value pairs.

THE PATTERN

```
<tag name="value">Content</tag>
```

lang Declares the language for screen readers.

href Provides the destination URL for a link.

src & alt Provides the image source and text alternative.

id & class Hooks for CSS styling and JavaScript.

```
<!-- Language declaration -->
<html lang="en">

<!-- Hypertext Reference (Link) -->
<a href="https://example.com">Visit Campus
Connect</a>

<!-- Source and Alternative Text -->


<!-- Identifiers for styling or scripts -->
<p id="intro-text"
class="highlight">Welcome!</p>
```

Nest in the order you open

Elements can be placed inside other elements, but they must be closed properly to maintain a valid document tree.

THE GOLDEN RULE

First In, Last Out. Close the most recently opened element before closing its parent.

- **Browsers guess your intent:** When tags overlap, the browser tries to fix it, often resulting in broken layouts.
- **Accessibility fails:** Screen readers rely on the document tree. Broken nesting destroys the structural map they use to navigate.

CORRECT

```
<p>  
  My cat is <strong>very</strong>  
  grumpy.  
</p>
```

BROKEN

```
<p>  
  My cat is <strong>very grumpy.</p>  
  </strong>
```

Headings create a content map

Headings (<h1> to <h6>) define the outline of your page. Screen readers use them to generate a table of contents for navigation.

■ One main title

Use exactly one <h1> per page to describe the overall topic.

■ Never skip levels

Do not jump from <h2> to <h4>. Headings are for structure, not for sizing text.

■ Paragraphs hold the text

Wrap regular text in <p> tags. Never use line breaks (
) to create spacing between blocks of text.

```
<!-- A logical document outline -->
<h1>Campus Connect Events</h1>
<p>Discover what is happening this week.</p>

<h2>Today's Workshops</h2>
<p>Join our technical sessions.</p>

  <h3>Intro to Web Development</h3>
  <p>Learn HTML basics in Room 101.</p>

  <h3>Advanced CSS Layouts</h3>
  <p>Master Flexbox in Room 102.</p>

<h2>Upcoming Career Fairs</h2>
<p>Meet top tech employers next month.</p>
```

Lists express related items

When content belongs together, wrap it in a list. This tells screen readers exactly how many items exist and how they relate to each other.

■ Unordered List

Use `<ul>` when the sequence of items does not matter.

```
<ul><li>Tech Workshop</li>
<li>Career Fair</li><li>Study
Group</li></ul>
```

BROWSER DEFAULT

- Tech Workshop
- Career Fair
- Study Group

■ Ordered List

Use `<ol>` when items follow a specific sequence or ranking.

```
<ol><li>Create account</li>
<li>Find an event</li>
<li>Register</li></ol>
```

BROWSER DEFAULT

1. Create account
2. Find an event
3. Register

■ Description List

Use `<dl>` for name/value pairs, like terms and definitions.

```
<dl><dt>Location</dt><dd>Room
402</dd><dt>Time</dt><dd>10:00
AM</dd></dl>
```

BROWSER DEFAULT

Location

Room 402

Time

10:00 AM

Links connect documents and actions

The anchor element (`<a>`) makes the web a web. It connects pages, files, and sections using the `href` attribute.

INACCESSIBLE (BAD)

To view the event schedule, [click here](#).

Screen reader users often list links out of context. "Click here" tells them nothing about the destination.

DESCRIPTIVE (GOOD)

View the [event schedule](#) for more details.

The link text itself describes exactly what the user will find when they follow it.

INTERNAL LINK (RELATIVE PATH)

```
<a href="/events/tech-workshop.html">
  View Tech Workshop
</a>
```

EXTERNAL LINK (ABSOLUTE PATH + SAFETY)

```
<a href="https://example.com"
  target="_blank"
  rel="noopener noreferrer">
  Visit Example Partner
</a>
```

FRAGMENT LINK (JUMP TO SECTION)

```
<a href="#registration-form">
  Jump to Registration
</a>
```

```
<h2 id="registration-form">Register Now</h2>
```

Semantic landmarks orient people

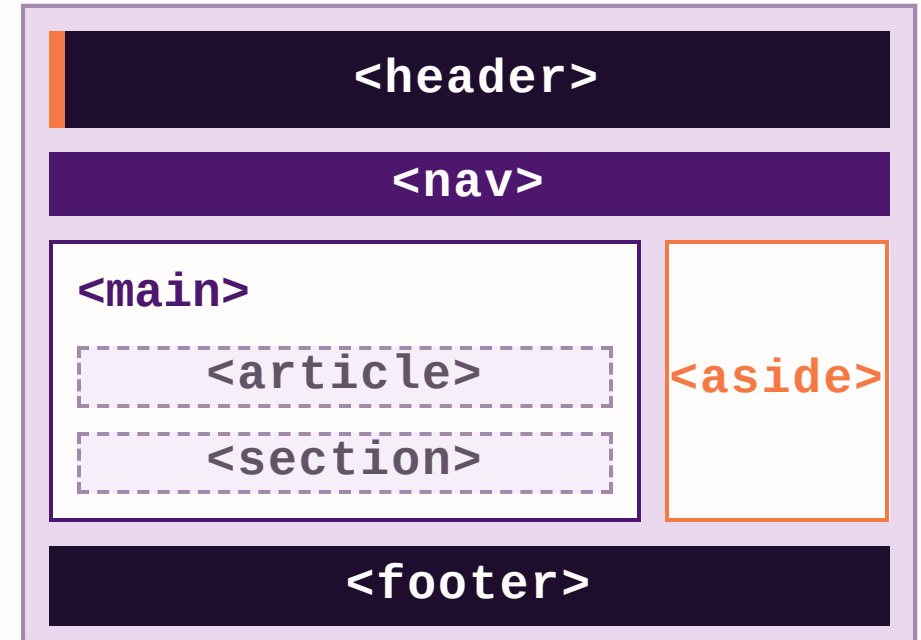
Landmarks divide a page into recognizable regions. They allow screen reader users to jump directly to the content they want, bypassing repetitive links.

THE PROBLEM WITH <DIV>

A `<div>` is a generic container with **zero semantic meaning**. If your entire page is built with divs, assistive technologies see a flat, unnavigable wall of text.

THE SEMANTIC SOLUTION

Tags like `<main>` and `<nav>` explicitly describe their purpose. A user can press a single key to skip the header and jump straight to the `<main>` content.



Build a complete first page

Bringing it all together. This is a complete, valid, and semantic HTML document for a Campus Connect student profile.

✓ Valid Document Skeleton

✓ Semantic Landmarks

✓ Logical Heading Hierarchy

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Student Profile | Campus Connect</title>
</head>
<body>
<header><h1>Campus Connect</h1>
<nav><ul>
<li><a href="#profile">My Profile</a></li>
<li><a href="#courses">My Courses</a></li>
</ul></nav></header>
<main>
<article id="profile"><h2>Alex Doe</h2><p>Software Engineering
Student</p></article>
<section id="courses"><h2>Enrolled Courses</h2><ul>
<li>Web Development Foundations</li>
<li>Database Systems</li>
</ul></section>
</main>
<footer><p>LEA Labs 2026</p></footer>
</body>
</html>
```

Foundation lab: build with meaning

THE BRIEF

Build a semantic "Student Profile" page for Campus Connect using **only HTML**. No CSS allowed.

DELIVERABLES

- Valid HTML5 document skeleton.
- Semantic `<header>` , `<main>` , `<footer>` . landmarks:
- Logical heading hierarchy (`<h1>` – `<h3>`) and lists.
- Descriptive links to external resources.

SUCCESS CRITERIA

- **Validation:** Passes the W3C HTML Validator.
- **Strict semantics:** No `<div>` or `<span>` where a semantic tag applies.
- **Hierarchy:** No skipped heading levels.
- **Nesting:** All elements properly closed.

EXTENSION CHALLENGE

Implement a concise `<dl>` for student metadata (Major, Grad Year, Advisor).